PHYSICAL CHEMISTRY CUMULATIVE EXAM

March 1, 1997

In photoelectron spectroscopy, the molecules are irradiated with monochromatic ultraviolet light. The kinetic energy of the electrons ejected is subsequently analyzed. The results may be plotted as electron beam intensity versus the kinetic energy of the electrons.

The photoelectron spectrum of cyclopropane is shown below (Basch et al., J. Chem. Phys. 1969), using He I irradiation (wave length 58.435 nm). The centers of three bands, corresponding to ionization from the e', e" and a_1 orbitals are shown by arrows.

1. What are the three lowest ionization potentials of cyclopropane (in eV)?

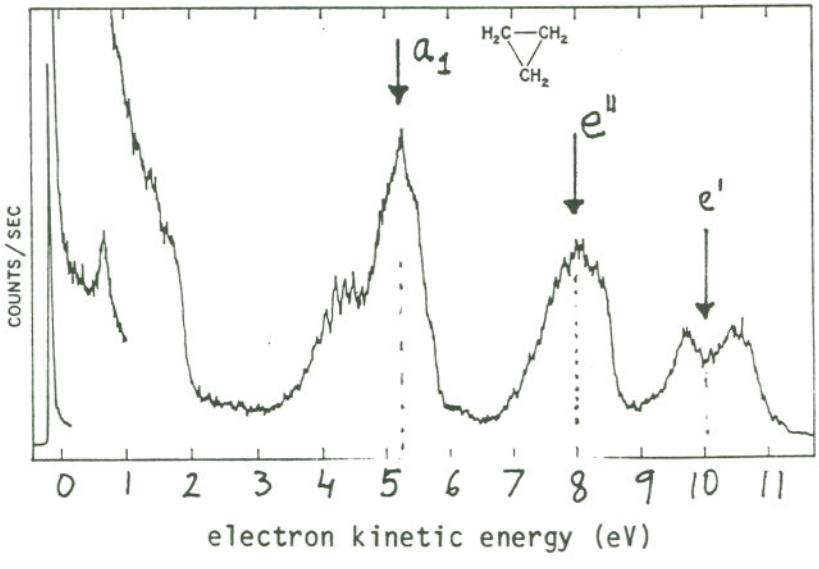
2. What are the terms symbols for the three ion states? Which of these states will undergo a Jahn-Teller distortion of the nuclear framework, and thus lose the full symmetry of cyclopropane?

3. The ground state of the ion is obtained by removing an electron from the highest occupied orbital, the e' one. The states formed by removing an electron from the lower (e" and a_1) orbitals are excited states of the ion and will spontaneously radiate a photon and decay to the ground ion state. What are the wave lengths of the radiation emitted in this process?

4. What is the symmetry point group of cyclopropane?

$c = 2.9979*10^{8}$ m/s (speed of light)

$h = 6.6262*10^{-34}$ Js (Planck's constant)

$1\ eV = 1.6020*10^{-19}$ J

Photoelectron spectrum of cyclopropane.